# THE SUPPORT OF A QUASI-COHERENT SHEAF IS CLOSED UNDER SPECIALIZATION

THE STACKS PROJECT AUTHORS

*Editorial note.* All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

*Editorial note.* Following the multi-claim convention for this source, source_url is the source-level URL https://stacks.math.columbia.edu/ and the source label lemma-support-closed-specialization is carried in claim_key and here rather than in a per-claim tag URL.

*Editorial note.* The preamble is the chapter's shared Stacks Project preamble, retained as sources/stacks-morphisms/retained-preamble.tex: a line-for-line excerpt of the shared Stacks preamble staged with this snapshot. No macro was invented or redefined, and this card adds no preamble text of its own.

*Editorial note.* Statement span 660-665 and proof span 667-672 come from the halfA pool card operations/pool-cards/stacks-morphisms-halfA.json and were re-read in the pinned chapter file before this card was written.

*Editorial note.* The proof is short by the author's own choice and is deliberately not padded; the formula text{Supp}(F) in the proof is the author's own notation.

*Allowed standard prerequisites (editorial).*

- Stalks of a sheaf of modules and the localization maps along specializations: for a specialization x' in the closure of {x} the stalk at x' of a module sheaf receives the stalk at x by localization at a multiplicative set. Used at proof lines 669-671, where F_{x'} is identified as a localization of the module F_x. Gives the reduction of the statement to the vanishing of a localization of a zero module.
- The support of F is the set of points with nonzero stalk, so its complement is the locus where the stalk vanishes. Used at statement line 663 and proof line 671. Turns the conclusion 'closed under specialization' into the equivalent statement that the complement is closed under generalization.

**Lemma 0.1.** *Let $X$ be a scheme. Let $\mathcal{F}$ be a quasi-coherent sheaf on $X$. The support of $\mathcal{F}$ is closed under specialization.*

*Proof.* If $x' \rightsquigarrow x$ is a specialization and $\mathcal{F}_x = 0$ then $\mathcal{F}_{x'}$ is zero, as $\mathcal{F}_{x'}$ is a localization of the module $\mathcal{F}_x$. Hence the complement of $\mathrm{Supp}(\mathcal{F})$ is closed under generalization. □

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